

OF-2 Report :

Simulating flow around a Circular Cylinder

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1 Introduction

Flow around a circular cylinder is a classic problem in *fluid mechanics* and is commonly studied in *computational fluid dynamics* (CFD) because of its interesting flow behavior at different Reynolds numbers (Re). At low Re , the flow is steady and symmetric, but as Re increases, the wake behind the cylinder becomes unsteady, first generating recirculation regions and ultimately leading to *vortex shedding* and the formation of a *von Kármán vortex street*. This type of flow is important in many engineering applications where understanding how vortices form and interact with surfaces can help improve designs.

The goal of this project is to **simulate flow past a circular cylinder using OpenFOAM** and analyze how the flow changes at different Reynolds numbers. Specifically, we will:

- **Generate and refine structured meshes** using `blockMesh` to see how grid resolution affects accuracy.
- **Simulate steady and unsteady flow** at $Re = 20$ (where flow remains steady) and $Re = 110$ (where vortex shedding occurs).
- **Analyze velocity, pressure, and forces**, including shear stress and drag.
- **Validate our results** by comparing vortex shedding frequency (Strouhal number) and separation point locations with published data from *Williamson (1988)* and *Park et al. (1998)*.

The simulations will be performed using the **OpenFOAM**, for *incompressible, laminar flow*. The computational setup includes **mesh generation, a no-slip boundary on the cylinder, and downstream data probes**. All variables are nondimensionalized using the cylinder diameter ($D = 1$ m) and freestream velocity ($U = 1$ m/s), which simplifies the Reynolds number to:

$$Re = \frac{UD}{\nu} = \frac{1}{\nu}. \quad (1)$$

In this report, we will go through the **mesh setup, simulation process, and post-processing results using ParaView** to understand how the flow behaves at different Reynolds numbers. By refining our meshes and analyzing key flow parameters, we aim to gain a better understanding of *how OpenFOAM handles external flow simulations and how mesh quality affects the results*.

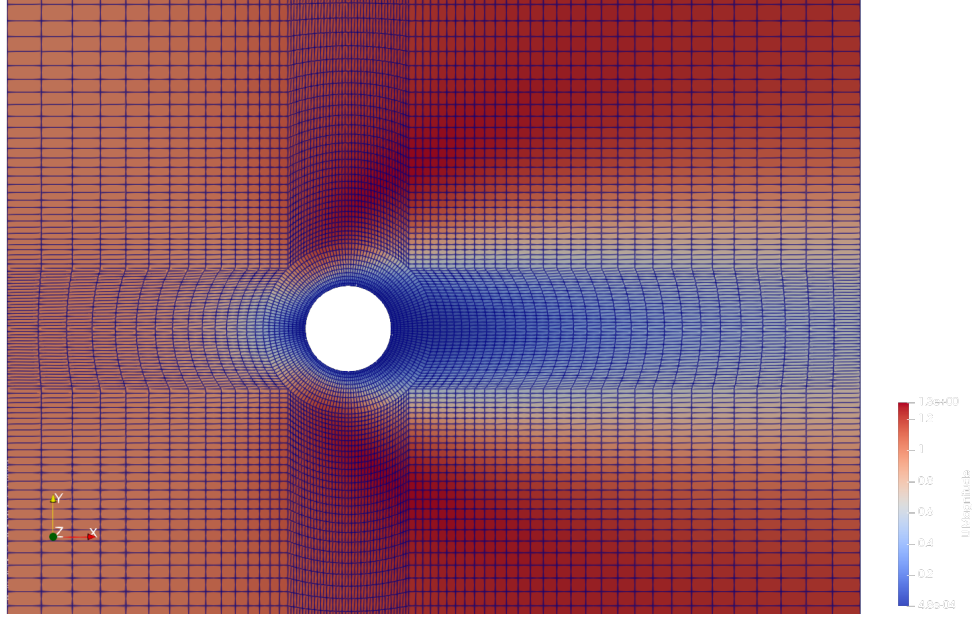


Figure 1: Visualization of the computational domain and mesh in ParaView.

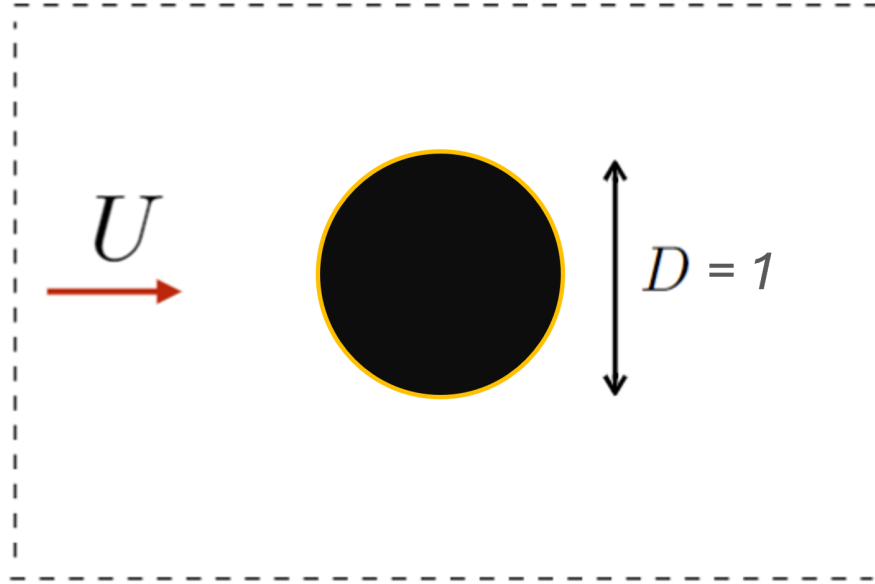


Figure 2: Structured mesh used for the simulation.

To illustrate the computational setup, Figure 1 shows the domain and mesh visualized in ParaView, and Figure 2 provides a detailed circular cylinder used in the study.

2 Assembly of preliminary meshes

2.1 Schematic of Mesh Generation

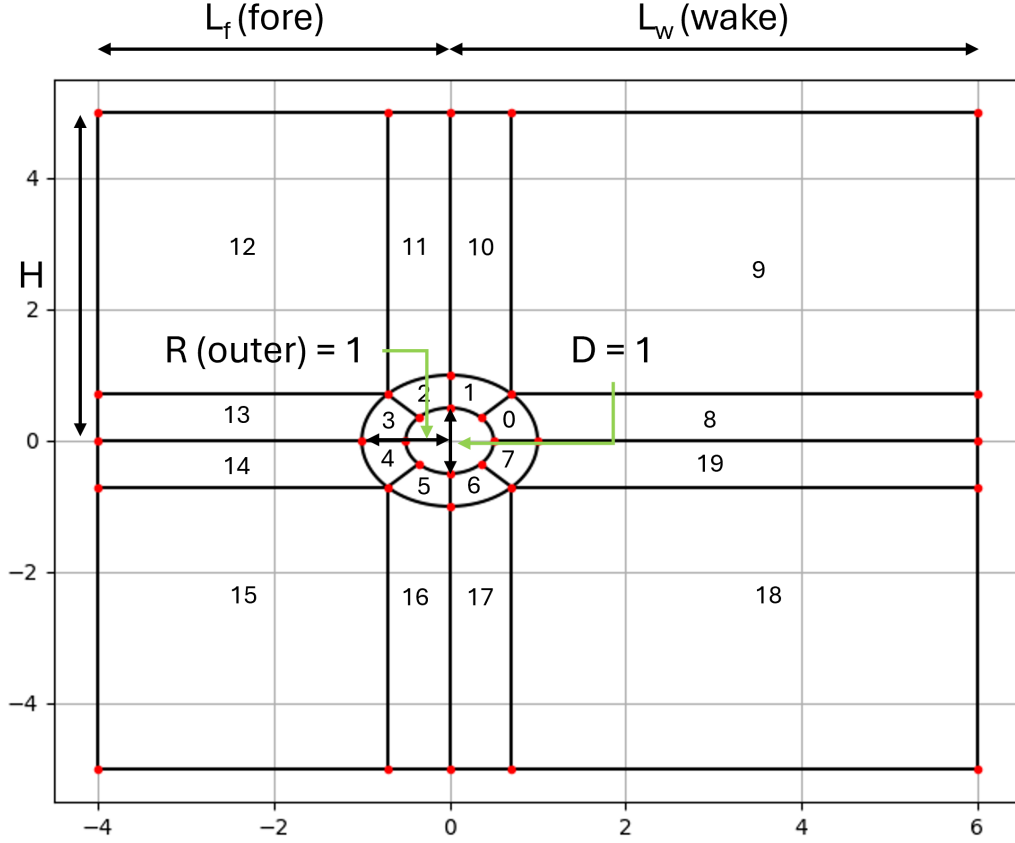


Figure 3: Generating the block mesh in $\mathbb{R}^2$

The block mesh is characterized by the fore length L_f , wake length L_w , half-height H , outer circle radius R , and the diameter of the circular cylinder D . These parameters can be changed within the code (found in the Appendix) to output new vertices, edges, and faces.

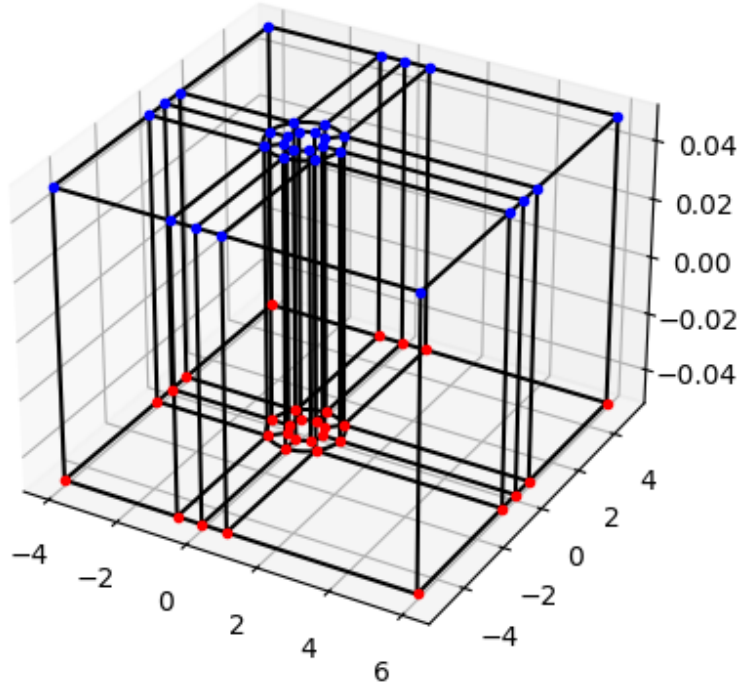


Figure 4: Visualization in $\mathbb{R}^3$

Since OPENFOAM is inherently a 3-dimensional solver, we generate our blocks in $\mathbb{R}^3$ within a very small z margin of -0.05 to 0.05 . This is the trick to keep it, for the most part, 2-dimensional. NOTE: Code included in the Appendix to reproduce meshes.

3 Solution for Re=20 and 110

3.1 Re=20

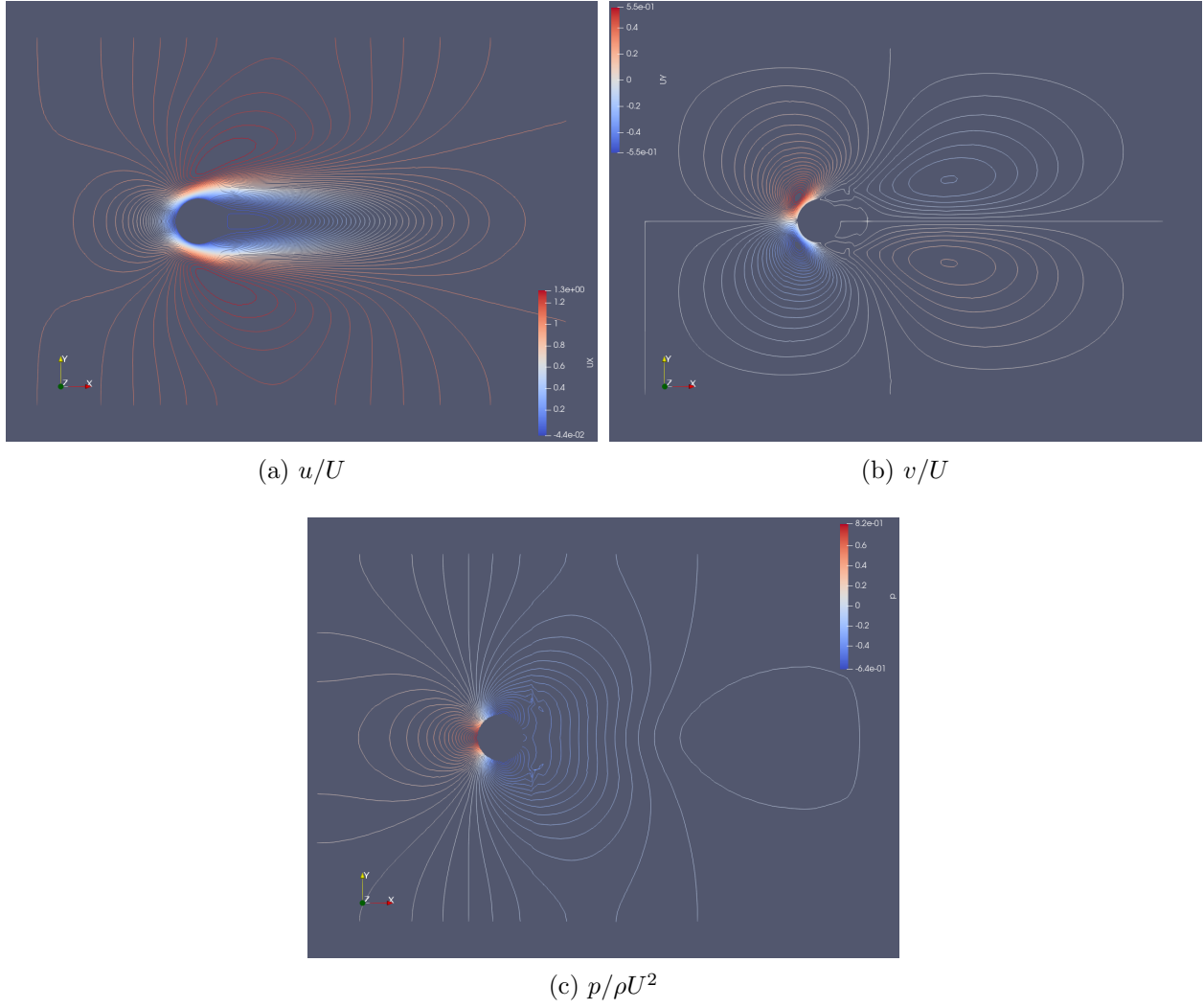


Figure 5: Velocity and pressure distributions

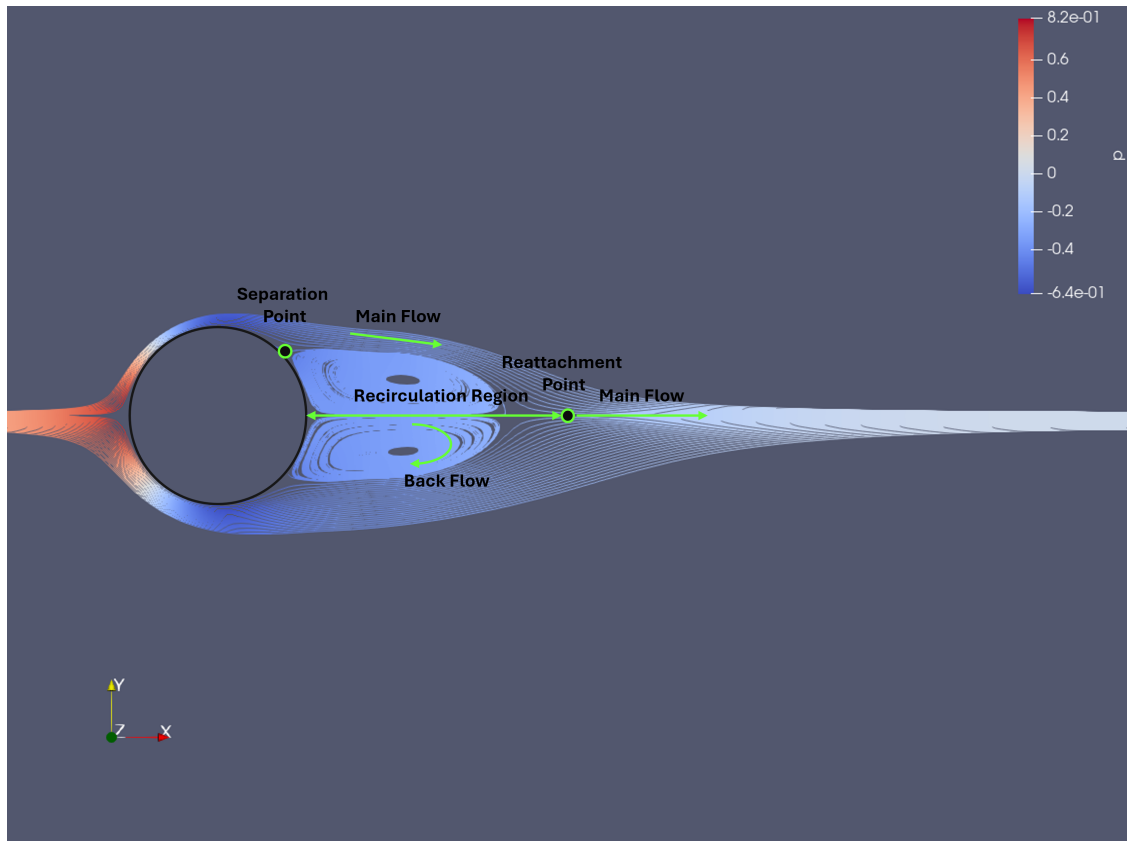


Figure 6: Visualizing streamlines for steady solution

3.2 Re=110

4 Improving the Mesh

5 Unsteady Flow: Strouhal Number of the vortex shedding

- 6 Steady Flow: velocity in the boundary layer, separation length, and rate of strain at the cylinder's wall

Appendix

A Code

PDF of code starts on next page.

Additionally, team repository: <https://github.com/andressuniaga/CFD-SP25-11>